

Human-Inspired Adaptive Optimal Control Framework for Robot-Environment Interaction

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Abstract—Enabling robots with uncertain dynamics to perform human-like adaptive operations in unknown environments remains a significant challenge in robotics research. Drawing inspiration from the dynamic modification of human arm muscles, we propose an innovative adaptive optimal control framework to address this issue. The framework integrates a variable optimal impedance adaptation (VOIA) method and an adaptive bias broad fuzzy neural network (ABBFNN) controller, facilitating adaptive manipulation behaviors in robot-environment interaction tasks. It can adaptively learn the impedance gain of unknown environment in the presence of uncertain robot dynamic model based on different task properties, simultaneously keeping the tracking error and interaction force optimized and minimized. The ABBFNN controller combines adaptive node increments to approximate uncertain dynamic model and introduces additional global bias and adaptive gain adjustment to improve the approximation accuracy and the rate of convergence significantly. VOIA seamlessly integrates a finely tuned proportional-integral-derivative (PID) variable target stiffness and an impact compensator, ensuring accurate responses to varying environmental conditions and improved disturbance rejection. Moreover, a momentum-based force observer is utilized within the framework for interaction force estimation, eliminating the need for force sensors and simplifying the system. Simulations and experiments validate the effectiveness and practicality of the proposed optimal interaction control framework, demonstrating its potential to propel robots toward interactions with unknown environments.

Index Terms—Adaptive fuzzy-logic control, adaptive optimal interaction, human-like muscle characteristic, impedance adaptation, unknown environment.

Received 19 November 2024; revised 10 March 2025; accepted 4 June 2025. Date of publication 24 June 2025; date of current version 20 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62303457 and Grant U21A20482, and in part by the Beijing Municipal Natural Science Foundation under Grant 4252053. This article was recommended by Associate Editor N. Sun. (Corresponding authors: Yuchuang Tong; Zhengtao Zhang.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TSMC.2025.3577296>.

Digital Object Identifier 10.1109/TSMC.2025.3577296

I. INTRODUCTION

IN VARIOUS fields, such as medical [1], industrial [2], and service [3], robots are involved in interacting with the environment to achieve human-like operation or collaboration. However, how robots achieve human-like safe and friendly operation under uncertain dynamic models, thereby optimizing behaviors to realize adaptive optimal interaction with unknown environments, remains a challenge in robotics research.

By emulating human behaviors, an interaction model can be constructed to minimize trajectory and force errors, thereby enabling adaptive adjustment of robot motion [4]. For example, in sphere target assembly task, success hinges on maintaining minimal radial position errors while exerting the least possible contact force to avoid component damage [5]. Moreover, when interacting with deformable objects, such as cable routing or sponge compression, it is imperative to minimize trajectory tracking errors while ensuring that the interaction force, which tends to increase with deformation, remains as low as possible, thereby safeguarding the object's integrity. In these applications, achieving low tracking error in conjunction with minimal force is critical for ensuring safe, stable interactions and facilitating rapid, responsive control [6].

Over the past few decades, researchers have dedicated substantial efforts to investigating the problem of robot interaction control and learning [7]. In general, when robot performs interaction tasks, its dynamic model often encounters inaccuracies and deviations due to the lack of precise parameters and measurements of the interaction, negatively impacting the control performance and task accuracy [8]. Therefore, achieving better model approximation is a core issue in dynamic control design. With the development of neural networks, many researchers have combined them with adaptive control methods to obtain uncertain models. Among various networks, radial basis function neural network (RBFNN) [9] is widely applied. Yang et al. [10] and Peng et al. [11] employed it to estimate uncertain dynamics and achieved torque control. Liu et al. [12] enhanced model approximation accuracy by introducing global/local biases into RBFNN. Furthermore, leveraging the excellent approximation capability of fuzzy-logic [13], the adaptive fuzzy neural network (FNN) has been widely studied [14], [15]. He and Dong [16] combined adaptive FNN control with impedance learning to control constrained robots. Ma et al. [17] presented an adaptive fuzzy control strategy identifying unknown nonlinearity in dynamics using fuzzy-logic systems. Li et al. [18] proposed an adaptive FNN command filtered impedance control method for estimating manipulator dynamics with constraints.

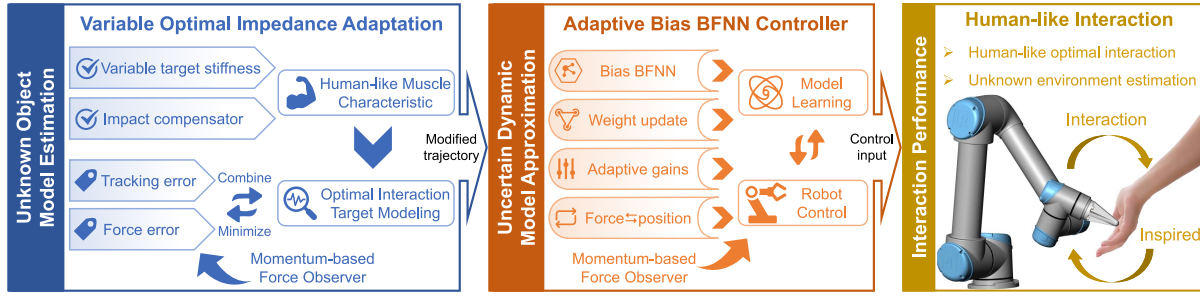


Fig. 1. Block diagram of human-inspired adaptive optimal interaction control framework.

However, the above control methods inevitably require prior parameters based on expert experience in the controller design. To reduce the impact of nodes, Huang et al. [19], [20] proposed a novel broad FNN (BFNN), combining the broad learning system (BLS) [21] with a FNN. This network utilizes the characteristic of BLS to dynamically select the number of nodes based on different tasks, reducing the need for manual parameter design and achieving a good approximation. However, while this method has improved significantly, expert experiences still influence its effectiveness.

In addition, humans can naturally modify the impedance and force to interact with environments. Biological studies suggest that humans can adjust the stiffness of muscles through the central nervous system, achieving smooth interaction [22]. This ability is a key to achieve safe, compliant, and adaptive physical interactions. For robots, impedance control dynamically adjusts the relationship between end-effector motion and force without decoupling, making them extensively applicable in complex interaction control tasks [23]. The critical factor in impedance control is the accurate acquisition of impedance parameters [24]. Typically, constructing an impedance model is vital for parameter acquisition. In some early works [25], [26], [27], fixed and artificially designed impedance models were employed and combined with a linear quadratic regulator (LQR) for interaction with a completely known environment. However, environment usually encompasses various unknown factors, such as unknown interaction objects with unstructured surfaces, unknown environment impedance models, and external noises and shocks, limiting the effectiveness of interaction control based on fixed impedance models. Although LQR can achieve optimal control under a vital precondition of completely known environment model, its effectiveness is significantly reduced in scenarios with high uncertainty, nonlinear dynamics, and unknown environment properties, which are typical of challenges in real-world robot-environment interaction. To address this issue, Ge et al. [28] extended the adaptive optimal control method [29] to robots, eventually obtaining the optimal stiffness through policy iteration learning to achieve optimal interaction in various unknown scenarios. Subsequently, Huang et al. [20] further improved it by incorporating a soft function to enhance control continuity. Roveda et al. [30] proposed an extended Kalman filter to estimate the stiffness of the environment and realize the optimal interaction. However, these methods usually use assumed

initial parameters and are challenging to avoid, resulting in interaction properties that may be affected by changes in the operating environment.

In response to the above challenges, we propose an adaptive optimal impedance control framework for robot-environment interaction control inspired by muscle characteristics to achieve human-like behavior in robot-environment interaction and realize versatility and adaptability for different interaction objects, as shown in Fig. 1. In terms of unknown environment model estimation, the variable optimal impedance adaptation (VOIA) method is presented to handle various unknown factors and achieve optimal interaction. Compared to LQR [25], [26] and impedance adaptation [20], [28], this method can estimate the environment model in real-time to avoid a priori unknown impedance and generate a smooth reference trajectory for interaction. Furthermore, VOIA effectively addresses unstructuredness caused by dynamic changes in the surface and physical properties of the object, and effectively mitigates impacts, such as noises and shocks. In terms of uncertain robot dynamic model approximation, an adaptive bias BFNN (ABBFNN) controller is designed to accomplish simultaneous learning and control. Compared to RBFNN [11], [12] and FNN [17], [18] controllers, the controller can adaptively adjust the control gains, eliminating the need for subjective manual adjustments of network parameters and nodes during the network design process, thus facilitating human-like adaptive and compliant operation.

Moreover, to comprehensively evaluate the integrated performance of the proposed framework compared with existing methods, Table I analyzes the aspects of environment model, robot dynamics, control signal, the way of network node design, parameter sensitivity, oscillation suppression capability, force sensor requirement, and real-world experiment, respectively. Overall, in terms of adaptability, the proposed framework requires minimal manually designed parameters and exhibits low sensitivity to parameter variations, making it highly effective for robot-environment interaction tasks with unknown and uncertain factors. In terms of oscillation suppression capability, VOIA and ABBFNN controller can effectively handle impacts during interaction and adaptively adjust robot behavior. In terms of hardware constraints, the presence of force observer simplifies the control system, and the framework considers the possibility that the robot system may not have joint torque or motor current interfaces, enhancing its compatibility with a wide

TABLE I
COMPARISONS AMONG DIFFERENT METHODS

	Unknown environment	Uncertain dynamics	Control signal	Network node	Parameter sensitivity	Oscillation suppression	Force sensor	Real-world experiment
Ours	Yes	Yes	Joint angular velocity	Adaptive	Low	Yes	No	Yes
[4]	Yes	Yes	Torque	Manual	High	No	Yes	No
[7]	Yes	Yes	Torque	Manual	Middle	No	Yes	No
[10]	No	Yes	Torque	Manual	Middle	No	Yes	Yes
[11], [20]	Yes	Yes	Torque	Adaptive	Middle	No	Yes	Yes
[15]	No	No	Torque	No	High	Yes	Yes	No
[16]	Yes	Yes	Torque	Manual	Middle	No	Yes	No
[17]	No	Yes	Torque	No	High	No	Yes	No
[18]	Yes	Yes	Torque	Manual	Middle	Yes	Yes	No
[19]	No	Yes	Torque	Adaptive	Middle	No	Yes	Yes
[28]	Yes	No	No	No	Low	No	Yes	No
[30]	Yes	No	Torque	No	Middle	No	No	Yes
[31], [32]	Yes	Yes	Torque	No	Low	Yes	Yes	Yes
[33]	Yes	Yes	Torque	Manual	Middle	No	No	Yes
[34]	Yes	No	Joint angular velocity	No	Low	Yes	Yes	Yes
[35]	No	Yes	Torque	Manual	Middle	Yes	Yes	Yes
[36], [37]	Yes	No	Torque	No	Low	Yes	Yes	Yes

range of robotic systems. Therefore, it can be concluded that the proposed framework demonstrates superior comprehensive performance, making it more practical for various robot-environment interaction tasks. The main contributions of this article are as follows.

- 1) Inspired by the characteristics of human arm muscles, this article presents an adaptive optimal interaction control framework for manipulator with uncertain dynamic models, facilitating safe and friendly interaction between manipulator and environment with unstructured surface, unknown impedance, and external noise and shocks.
- 2) The ABBFNN controller combines global bias and adaptive control gain adjustment, which enables accurate approximation of uncertain dynamic model and drives the manipulator to interact with the environment.
- 3) The VOIA method integrates variable target stiffness and impact compensator, which can accurately respond to unknown environment, estimate impedance gain, and modify the trajectory.
- 4) The sensorless momentum-based force observer estimates interaction force while mitigating signal interference during data collection, simplifying the control system.

The remainder of this article is organized as follows. Section II presents VOIA. Section III proposes the ABBFNN controller. Section IV devises the adaptive optimal control framework. Section V verifies the effectiveness through simulations and real-world experiments. Section VI provides the conclusion.

II. VARIABLE OPTIMAL IMPEDANCE ADAPTATION

In this section, we introduce the dynamic model of robot-environment interaction and the variable optimal impedance adaptive method to mimic human muscle characteristic.

A. Dynamic Model of Robot-Environment Interaction

The forward kinematics model in velocity level of the manipulator is

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$$

where $\mathbf{x} \in \mathcal{R}^m$ is the end-effector position, $\mathbf{q} \in \mathcal{R}^n (m \leq n)$ is the joint angles of the manipulator, $\mathbf{J}(\mathbf{q}) \in \mathcal{R}^{m \times n}$ is the Jacobin matrix of the manipulator, in which m and n are the dimension of the task space and joint space, respectively.

When the manipulator interacts with the environment, the dynamic model in joint space [38] can be expressed as

$$\mathbf{M}_q(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{V}_q(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}_q(\mathbf{q}) + \boldsymbol{\tau}_e = \boldsymbol{\tau}_c \quad (1)$$

where $\mathbf{M}_q(\mathbf{q}) \in \mathcal{R}^{n \times n}$ is the inertia matrix in joint space, $\mathbf{V}_q(\mathbf{q}, \dot{\mathbf{q}}) \in \mathcal{R}^{n \times n}$ is the Coriolis and centrifugal matrix, $\mathbf{g}_q(\mathbf{q}) \in \mathcal{R}^n$ is the gravity vector, $\boldsymbol{\tau}_c \in \mathcal{R}^n$ is the joint control torque of the manipulator, $\boldsymbol{\tau}_e = \mathbf{J}(\mathbf{q})^T \mathbf{f}_e \in \mathcal{R}^n$ is the external torque interacting with the environment, in which $\mathbf{f}_e$ is the interaction force. Furthermore, The dynamic model in task space can be written as

$$\mathbf{M}_x(\mathbf{x})\ddot{\mathbf{x}} + \mathbf{V}_x(\mathbf{x}, \dot{\mathbf{x}})\dot{\mathbf{x}} + \mathbf{g}_x(\mathbf{x}) + \mathbf{f}_e = \mathbf{J}^T \boldsymbol{\tau}_c$$

where $\mathbf{M}_x(\mathbf{x}) \in \mathcal{R}^{m \times m}$, $\mathbf{V}_x(\mathbf{x}, \dot{\mathbf{x}}) \in \mathcal{R}^{m \times m}$ and $\mathbf{g}_x(\mathbf{x}) \in \mathcal{R}^m$ are defined as

$$\begin{aligned} \mathbf{M}_x(\mathbf{x}) &= \mathbf{J}^T \mathbf{M}_q(\mathbf{q}) \mathbf{J}^{-1} \\ \mathbf{V}_x(\mathbf{x}, \dot{\mathbf{x}}) &= \mathbf{J}^T (\mathbf{V}_q(\mathbf{q}, \dot{\mathbf{q}}) - \mathbf{M}_q(\mathbf{q}) \mathbf{J}^{-1} \dot{\mathbf{J}}) \mathbf{J}^{-1} \\ \mathbf{g}_x(\mathbf{x}) &= \mathbf{J}^T \mathbf{g}_q(\mathbf{q}). \end{aligned}$$

However, the assumption of precise dynamic parameters is practically unrealistic due to the lack of precise parameters and measurements of the interaction between the robot and environment. This uncertainty, stemming from discrepancies between the ideal mathematical model and the actual physical system, can negatively affect control performance, such as tracking accuracy and repeatability.

Moreover, the target impedance function of interaction in the task space [28] can be expressed as

$$\mathbf{B}_e \dot{\mathbf{e}}_x + \mathbf{K}_e \mathbf{e}_x = -\mathbf{f}_e \quad (2)$$

where $\mathbf{B}_e, \mathbf{K}_e$ are the damping and stiffness matrices, respectively; $\mathbf{e}_x = \mathbf{x}_r - \mathbf{x}_d \in \mathcal{R}^m$ is the position error, in which $\mathbf{x}_r$ is the reference trajectory and $\mathbf{x}_d$ is the desired trajectory. In the absence of interaction, the end-effector tracks the desired trajectory exactly, i.e., $\mathbf{x} = \mathbf{x}_r = \mathbf{x}_d$, such that $\mathbf{f}_e = 0$.

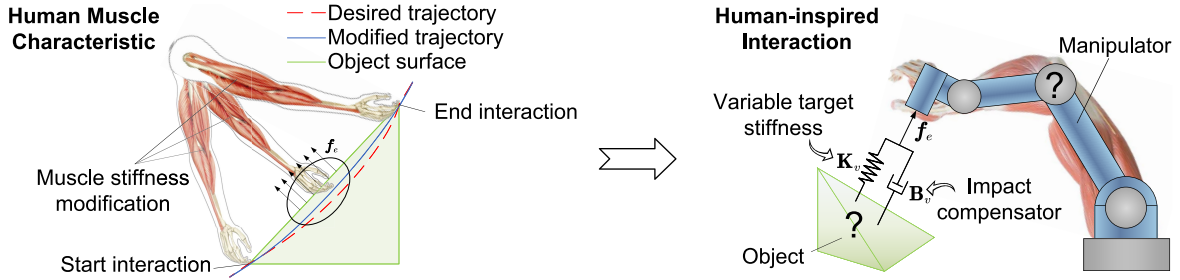


Fig. 2. Variable impedance modeling method to mimic human arm muscle.

B. Variable Impedance Modeling

As shown in Fig. 2, humans can swiftly adapt to changing environments, adjusting actions and interactions in real-time. Inspired by this characteristic, we propose an variable impedance modeling method to simulate the ability of the human neuromuscular system to adjust stiffness and damping based on task requirements and environmental conditions.

First, we introduce a state variable $\mathbf{y} = [\mathbf{x}^T, \mathbf{s}^T]^T$, where $\mathbf{s} \in \mathcal{R}^k$ serves as an intermediate state variable defined as

$$\begin{cases} \dot{\mathbf{s}} = \mathbf{A}\mathbf{s} \\ \mathbf{x}_d = \mathbf{E}\mathbf{s} \end{cases}$$

where $\mathbf{A} \in \mathcal{R}^{k \times k}$ and $\mathbf{E} \in \mathcal{R}^{m \times k}$ are two adjustable matrices.

Next, (2) can be rewritten as a linear state space system

$$\dot{\mathbf{y}} = \mathbf{C}\mathbf{y} + \mathbf{D}\mathbf{f}_e \quad (3)$$

where $\mathbf{C} = \text{diag}\{-\mathbf{B}_e\mathbf{K}_e^{-1}, \mathbf{A}\}$, $\mathbf{D} = [-\mathbf{B}_e^{-1}, \mathbf{0}]^T$.

To mimic the muscle stiffness modification process, a variable target stiffness $\mathbf{K}_v$, updating $\mathbf{K}_e$ in (2) by force/position feedback and extends the force error in a proportional-integral-derivative (PID) manner [34], is defined as

$$\mathbf{K}_v = -\frac{k_p \mathbf{e}_f + k_i \int_{t_i}^{t_{i+1}} \mathbf{e}_f dt + k_d \dot{\mathbf{e}}_f}{\mathbf{e}_x} \quad (4)$$

where k_p , k_i , and k_d are the proportional, integral, and differential gains, respectively; $\mathbf{e}_f = \mathbf{f}_e - \mathbf{f}_d$ is the force error, in which $\mathbf{f}_d$ is the desired force.

In this article, $\mathbf{f}_d$ is set to 0 to promote impedance learning [20], thereby fostering human-like behavior and optimal interaction performance. This setting enables the system to rapidly correct deviations from stability and prevent the excessive forces, ensuring safe and stable robot-environment interaction.

Simultaneously, we incorporate a variable damping coefficient $\mathbf{B}_v$ in place of $\mathbf{B}_e$ in (2) as a simple impact compensator to avoid oscillations, defined as

$$\mathbf{B}_v = \begin{cases} \mathbf{B}_e, & \delta^{\text{Imp}} \leq \eta_1 \\ \mathbf{B}_1 \tanh(\alpha) + \mathbf{B}_2, & \delta^{\text{Imp}} > \eta_1 \end{cases} \quad (5)$$

$$\alpha = \delta^{\text{Imp}} / \eta_2$$

where $\delta^{\text{Imp}} = \int_{t_i}^{t_{i+1}} |\mathbf{f}_e - \mathbf{f}_d| dt$ is an coefficient designed to determine whether the variable damping coefficient is activated, $\mathbf{B}_1$ and $\mathbf{B}_2$ are damping parameters, η_1 and η_2 are obtained by the trial-and-error method, $\tanh(\alpha) = (e^\alpha - e^{-\alpha}) / (e^\alpha + e^{-\alpha})$ is for reducing the effects of noises and shocks.

Thus, the coefficient matrices of (3) can be transformed into $\mathbf{C} = \text{diag}\{-\mathbf{B}_v\mathbf{K}_v^{-1}, \mathbf{A}\}$, $\mathbf{D} = [-\mathbf{B}_v^{-1}, \mathbf{0}]^T$. Then, to achieve better interaction performance by minimizing the combination of tracking error and force error, a variable impedance model is defined as

$$\begin{aligned} \mathfrak{J} &= \int_0^\infty [(\mathbf{x}_r - \mathbf{x}_d)^T \mathbf{N} (\mathbf{x}_r - \mathbf{x}_d) \\ &\quad + (\mathbf{f} - \mathbf{f}_d)^T \mathbf{H} (\mathbf{f} - \mathbf{f}_d)] dt \\ &= \int_0^\infty [\mathbf{y}^T \mathbf{N}' \mathbf{y} + \mathbf{f}^T \mathbf{H} \mathbf{f}] dt \end{aligned} \quad (6)$$

where $\mathbf{f}$ is the interaction force, $\mathbf{N} \in \mathcal{R}^{m \times m}$ is the tracking error weight, $\mathbf{H} \in \mathcal{R}^{m \times m}$ is the interaction force weight, and $\mathbf{N}'$ is defined as

$$\mathbf{N}' = \begin{bmatrix} \mathbf{N} & -\mathbf{N}\mathbf{E} \\ -\mathbf{E}^T \mathbf{N} & \mathbf{E}^T \mathbf{N} \mathbf{E} \end{bmatrix}.$$

Remark 1: The proposed variable impedance modeling method reflects a similar natural adaptability of human muscles to adjust impedance by a PID-based variable target stiffness, which dynamically adjusts stiffness based on feedback. Furthermore, the compensatory mechanism of impact compensator can respond to sudden impacts as humans do, which helps the robot optimize behaviors and significantly reduce oscillations to ensure that the robot can maintain safe and friendly interactions even in unpredictable environments.

C. Adaptive Optimal Impedance Learning

Next, adaptive optimal impedance learning is employed to solve for the optimal feedback gain $\mathbf{G}^*$ and reference trajectory $\mathbf{x}_r$, as shown in Algorithm 1.

For convenience, the following matrix is designed as

$$\begin{aligned} \bar{\mathbf{y}} &= [\mathbf{y}_1^2, \mathbf{y}_1 \mathbf{y}_2, \dots, \mathbf{y}_1 \mathbf{y}_m, \mathbf{y}_2^2, \dots, \mathbf{y}_{m-1} \mathbf{y}_m, \mathbf{y}_m^2]^T \\ \Delta \bar{\mathbf{y}} &= [\bar{\mathbf{y}}(t_1) - \bar{\mathbf{y}}(t_0), \dots, \bar{\mathbf{y}}(t_l) - \bar{\mathbf{y}}(t_{l-1})]^T \\ \mathbf{I}_{yy} &= \left[\int_{t_0}^{t_1} \mathbf{y} \otimes \mathbf{y} dt, \dots, \int_{t_{l-1}}^{t_l} \mathbf{y} \otimes \mathbf{y} dt \right]^T \\ \mathbf{I}_{yf} &= \left[\int_{t_0}^{t_1} \mathbf{y} \otimes \mathbf{f} dt, \dots, \int_{t_{l-1}}^{t_l} \mathbf{y} \otimes \mathbf{f} dt \right]^T \\ \hat{\mathbf{P}}_i &= [\mathbf{P}_{1,1}, 2\mathbf{P}_{1,2}, \dots, 2\mathbf{P}_{1,m}, \mathbf{P}_{2,2}, 2\mathbf{P}_{2,3}, \dots, \\ &\quad 2\mathbf{P}_{m,m-1}, \mathbf{P}_{m,m}] \end{aligned} \quad (7)$$

where “ $\otimes$ ” represents the Kronecker product, $\mathbf{f} = \mathbf{G}_0 \mathbf{y}$ is the virtual interaction force, l is the length of time.

Algorithm 1 VOIA

Input: Desired end-effector trajectory $\mathbf{x}_d$, running time T , PID gains k_p , k_i , and k_d , interaction force $\mathbf{f}_e$, exploration noise $\boldsymbol{\varepsilon}$, initial position $\mathbf{y}_0$, adjustable matrices $\mathbf{A}$, $\mathbf{E}$, damping matrix $\mathbf{B}_e$, damping parameters $\mathbf{B}_1$ and $\mathbf{B}_2$, impact evaluation parameters η_1 and η_2 , iteration stopping threshold ϵ , gain adjustment period T_{12} .

Output: Optimized gain $\mathbf{G}^*$ and reference trajectory $\mathbf{x}_r$.

Step 1:

while $\text{rank}[\mathbf{I}_{yy}, \mathbf{I}_{yf_e}] < \frac{(m+k)(m+k+1)}{2} + m(m+k)$ **do**
 Calculate the time-varying target stiffness coefficient $\mathbf{K}_v$
 and damping coefficient $\mathbf{B}_v$
 Employ $\mathbf{f}_0 = \mathbf{G}_0 \mathbf{y} + \boldsymbol{\varepsilon}$ as the input interaction force
 Compute the operators $\Delta_{\bar{y}}, \mathbf{I}_{yy}, \mathbf{I}_{yf_e}$

end while

Step 2:

while $\|\hat{\mathbf{P}}_i - \hat{\mathbf{P}}_{i-1}\| > \epsilon$ **do**

 Solve $\mathbf{P}_i$ and $\mathbf{G}_{i+1}$ from (8)

end while

Adjust the impedance gain $\mathbf{G}$ from $\mathbf{G}_0$ to $\mathbf{G}^*$ during the adjustment period T_{12}

Step 3:

while $t \leq T$ **do**

 Use $\mathbf{f} = \mathbf{G}^* \mathbf{y}$ as the approximate optimal interaction force of the input

 Calculate the reference trajectory $\mathbf{x}_r$ by (3)

end while

Then, (6) can be rewritten as a system of linear equation in the matrix form [29] as

$$\begin{bmatrix} \hat{\mathbf{P}}_i \\ \text{vec}(\mathbf{G}_{i+1}) \end{bmatrix} = (\Lambda_i^T \Lambda_i)^{-1} \Lambda_i^T \Upsilon_i \quad (8)$$

where $\Lambda_i = [\Delta_{\bar{y}}, -2\mathbf{I}_{yy}(\mathbf{I}_m \otimes \mathbf{G}_i^T \mathbf{H}) - 2\mathbf{I}_{yf_e}(\mathbf{I}_m \otimes \mathbf{H})]$, in which $\mathbf{I}_m$ is a m -dimensional unit matrix; $\Upsilon_i = -\mathbf{I}_{yy} \text{vec}(\mathbf{N}_i)$, in which $\text{vec}(\cdot)$ is an operator used to stretch $\mathcal{R}^{a \times b}$ to $\mathcal{R}^{1 \times ab}$; $\mathbf{N}_i = \mathbf{N} + \mathbf{G}_i^T \mathbf{H} \mathbf{G}_i^T$.

To improve the smoothness of the reference trajectory, we utilize “sin” to introduce an auxiliary soft function defined as

$$\mathbf{G}_t = \frac{\mathbf{G}^* + \mathbf{G}_0}{2} + \frac{\mathbf{G}^* - \mathbf{G}_0}{2} \sin\left(-\frac{\pi}{2} + \pi \frac{t}{T_{12}}\right) \quad (9)$$

where T_{12} is the gain adjustment period.

Remark 2: Unlike relevant and optimal baseline LQR [25], [26] that require complete prior knowledge of the environment, our method operates without priors and estimates the unknown environmental impedance model. Compared to impedance adaptation in [28] and [20] that directly adopts the initial assumed environment model, our method introduces variable target stiffness to adjust stiffness coefficient in real-time based on motion and force feedback, enhancing adaptability to unstructured environments with dynamic changes. Furthermore, the impact compensator effectively prevents noises and shocks.

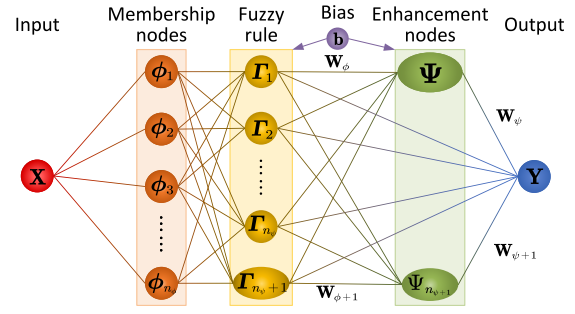


Fig. 3. Structure of bias BFNN.

III. ADAPTIVE BIAS BFNN CONTROLLER

In this section, we propose an ABBFNN controller combining bias BFNN with adaptive control gains for uncertain dynamic model estimation and interaction control.

A. Bias Broad Fuzzy Neural Network

In this article, we improve BFNN [20] by introduce a global bias term to accurately approximate uncertain robot dynamics, thereby optimizing its performance in complex trajectory tracking control tasks. The structure of bias BFNN is shown in Fig. 3.

To make BLS more applicable for robot control, we select the transformation function to be Gaussian function as

$$\phi_i(\mathbf{q}) = \exp\left[\frac{-(\mathbf{q} - \boldsymbol{\mu}_i)^T(\mathbf{q} - \boldsymbol{\mu}_i)}{\sigma^2}\right], \quad i = 1, 2, \dots, n_\phi$$

where $\boldsymbol{\mu}_i = [\mu_{i,1}, \mu_{i,2}, \dots, \mu_{i,n}]^T \in \mathcal{R}^n$ is the center of the Gaussian function, σ is the standard deviation, and n_ϕ is number of membership nodes.

In BFNN, the initial number of membership nodes n_ϕ is initially set to 1. This value increases adaptively based on the Euclidean distance between the reference trajectory and the center of the Gaussian function. The distance is denoted as

$$d(\mathbf{q}, \bar{\boldsymbol{\mu}}_{\min}) = \|\mathbf{q} - \bar{\boldsymbol{\mu}}_{\min}\| \quad (10)$$

where $\bar{\boldsymbol{\mu}}_{\min} = (\sum_{j=1}^{n_{\min}} \bar{\boldsymbol{\mu}}_{\min,j})/n_{\min}$ is the average of the first $n_{\min}$ centers in $\boldsymbol{\mu}_i$ closest to $\mathbf{q}$. A new membership node is added when the distance d exceeds the threshold $\iota > 0$, i.e., $d > \iota$. The center of the new node is defined as

$$\boldsymbol{\mu}_{i,n_\phi+1} = \bar{\boldsymbol{\mu}}_{\min} + \kappa(\mathbf{q} - \bar{\boldsymbol{\mu}}_{\min}) \quad (11)$$

where κ is the center update step.

Next, a first-order Takagi-Sugeno (TS) fuzzy rule [39] is applied. The fire strength of j th fuzzy rule is calculated as

$$\Gamma_j = \prod_{k=1}^{\varpi} \phi_{k,j}(\mathbf{q}), \quad j = 1, 2, \dots, n_\psi$$

where ϖ is the dimension of each sample, n_ψ is the number of enhancement nodes.

The transformation function Ψ_j corresponding to the enhancement nodes utilize the trigonometric expansion scheme, which reduces the computational burden, defined as

$$\Psi_j = [\cos(\Gamma_j), \sin(\Gamma_j)], \quad j = 1, 2, \dots, n_\psi.$$

As a result, the enhancement nodes matrix becomes

$$\Psi = [\Psi_1, \dots, \Psi_{n_\psi}] \\ = [\cos(\Gamma_1), \sin(\Gamma_1), \dots, \cos(\Gamma_{n_\psi}), \sin(\Gamma_{n_\psi})] \in \mathcal{R}^{n_x \times 2n_\psi}$$

where n_x is the dimension of the input $\mathbf{X}$. Consequently, the output of the network is expanded as

$$\mathbf{Y} = [\Gamma_1, \Gamma_2, \dots, \Gamma_{n_\psi} \mid \Psi_1, \Psi_2, \dots, \Psi_{n_\psi}] \mathbf{W} \\ = [\Gamma \mid \Psi] \mathbf{W} \\ = \sqsupset \mathbf{W}$$

where $\sqsupset \in \mathcal{R}^{n_x \times 3n_\psi}$ is the outcome of extension through enhancement nodes, $\mathbf{W} \in \mathcal{R}^{3n_\psi \times n_y}$ is the weight matrix of the network, in which n_y is the dimension of the output $\mathbf{Y}$.

Remark 3: Unlike BFNN [20], which may fall into local minima and lead to weakened control performance, our method introduces a global bias term in $\sqsupset$ as

$$\sqsupset = [\Gamma \mid \Psi] \\ = [\Gamma_1, \Gamma_2, \dots, \Gamma_{n_\psi} \mid \Psi_1, \Psi_2, \dots, \Psi_{n_\psi}] + \mathbf{b} \quad (12)$$

where $\mathbf{b} \in \mathcal{R}^{n_x \times 3n_\psi}$ is the bias matrix of the transformation functions. Including bias enables the BFNN to achieve higher accuracy in dynamic model estimation, faster adaptability to environmental changes, and an increased convergence speed, thereby enhancing the tracking performance.

After adding a new node, the updated node matrix becomes $\sqsupset_{\text{new}} = [\sqsupset \mid \Psi_{n_\psi+1}] \in \mathcal{R}^{n_x \times 3(n_\psi+1)}$, the connection weight matrix is expanded as $\mathbf{W}_{\text{new}} = [\mathbf{W}_{\text{old}}^T \mid \mathbf{0}]^T$, where $\mathbf{W}_{\text{old}}$ is the old weight matrix and $\mathbf{0} \in \mathcal{R}^{n_x \times 3}$. Therefore, once a new node is added, the output of bias BFNN can be denoted as

$$\mathbf{Y} = \sqsupset_{\text{new}} \mathbf{W}_{\text{new}} \\ = [\Gamma_{\text{new}} \mid \Psi_{\text{new}}] \mathbf{W}_{\text{new}} \\ = ([\Gamma_{\text{old}}, \Gamma_{n_\psi+1} \mid \Psi_{\text{old}}, \Psi_{n_\psi+1}] + \mathbf{b}) \begin{pmatrix} \mathbf{W}_{\text{old}} \\ \mathbf{W}_{\phi\psi} \end{pmatrix}$$

where $\Gamma_{\text{old}} = [\Gamma_1, \dots, \Gamma_{n_\psi}]$ and $\Psi_{\text{old}} = [\Psi_1, \dots, \Psi_{n_\psi}]$ are the fuzzy rules matrix and enhancement nodes matrix before extension, respectively; $\mathbf{W}_{\phi\psi} = [\mathbf{W}_{\phi+1}^T \mid \mathbf{W}_{\psi+1}^T]^T$ is the new weight matrix after node addition.

B. ABBFNN Controller Construction

Further, an ABBFNN controller is constructed using bias BFNN and adaptive gain adjustment. Each step of the controller is detailed in Algorithm 2.

First, we define a virtual control term $\hat{h}$ as

$$\hat{h} = \dot{q}_r + \mathbf{K}_q \mathcal{E}$$

where $q_r \in \mathcal{R}^n$ is the reference trajectory in joint space, $\mathbf{K}_q = \text{diag}\{k_{q,1}, \dots, k_{q,n}\} \in \mathcal{R}^n$ is a control gain, $\mathcal{E} \in \mathcal{R}^n$ is defined by L'Hospital's rule as

$$\mathcal{E} = [\varepsilon_1, \dots, \varepsilon_n]^T \\ = \left[\frac{\eta_{q,1}^2}{2\pi e_{q,1}} \sin\left(\frac{\pi e_{q,1}^2}{\eta_{q,1}^2}\right), \dots, \frac{\eta_{q,n}^2}{2\pi e_{q,n}} \sin\left(\frac{\pi e_{q,n}^2}{\eta_{q,n}^2}\right) \right]^T$$

where $e_q = q_r - q$ is the tracking error, $\eta_q \in \mathcal{R}^n$ is a constant associated with trajectory tracking error.

Algorithm 2 ABBFNN Controller

Input: Reference trajectory x_r , running time T , the number of centers $n_{\min}$, distance threshold ι , center point updated rate κ , bias $\mathbf{b}$, control parameters $\mathbf{K}_{q,0}$, $\mathbf{K}_{\dot{q},0}$, χ , ρ , learning rate ι_M , ι_V , ι_g , ν_M , ν_V , ν_g , weight update threshold η_ℓ .

Output: Joint space optimized trajectory q^* .

while $t \leq T$ **do**

Calculate joint reference trajectory $q_r, \dot{q}_r, \ddot{q}_r$ and virtual control term $\hat{h}$ by combining $x_r, \dot{x}_r, \ddot{x}_r$ and $q_a, \dot{q}_a, \ddot{q}_a$
Compute $d(q, \bar{\mu}_{\min})$ and determine whether to add nodes
Update transformation function matrix $\sqsupset$ and connection weight $\mathbf{W}$

Generate weight updated law $\hat{\mathbf{W}}_M, \hat{\mathbf{W}}_V, \hat{\mathbf{W}}_g$

Calculate weight $\hat{\mathbf{W}}$ and control torque τ_c by ABBFNN

Transform τ_c to $q, \dot{q}, \ddot{q}$

end while

Next, the interaction force is incorporated into the adaptive optimal interaction controller to achieve optimal interaction control between the robot and the unknown environment. The robot control torque can be defined as

$$\tau_c = \hat{\mathbf{M}}_q \dot{\hat{h}} + \hat{\mathbf{V}}_q \hat{h} + \hat{\mathbf{g}}_q + \mathbf{K}_{\dot{q}} J + e_q + J^T f_e \quad (13)$$

where $J = \dot{h} - \dot{q}$, $\mathbf{K}_{\dot{q}} = \text{diag}\{k_{\dot{q},1}, \dots, k_{\dot{q},n}\} \in \mathcal{R}^n$ is also a control gain similar to $\mathbf{K}_{\dot{q}}$.

In the improved ABBFNN controller, the control gains $\mathbf{K}_q$ and $\mathbf{K}_{\dot{q}}$ incorporate a soft function corresponding to the error term $e_q, e_{\dot{q}}$, defined as

$$\mathbf{K}_q = \mathbf{K}_{q,0} + \tanh\left(\frac{e_q}{\chi}\right) \rho \\ \mathbf{K}_{\dot{q}} = \mathbf{K}_{\dot{q},0} + \tanh\left(\frac{e_{\dot{q}}}{\chi}\right) \rho \quad (14)$$

where $\mathbf{K}_{q,0}$ and $\mathbf{K}_{\dot{q},0}$ are randomized initial gain matrices, $\chi \in \mathcal{R}^n$ is the gain adjustment threshold based on the impact of each joint on end-effector position, ρ is the adjustment step.

Furthermore, $\hat{\mathbf{M}}_q$, $\hat{\mathbf{V}}_q$, and $\hat{\mathbf{g}}_q$ in (13) are the estimation results of the uncertain robot dynamic model (1) by bias BFNN with the output

$$Y = [\hat{\mathbf{M}}_q, \hat{\mathbf{V}}_q, \hat{\mathbf{g}}_q] = \sqsupset \hat{\mathbf{W}} + \zeta \quad (15)$$

where $\zeta = [\zeta_M, \zeta_V, \zeta_g]$ is the approximation error matrix, $\sqsupset = [\sqsupset_M, \sqsupset_V, \sqsupset_g]$ and $\hat{\mathbf{W}} = [\hat{\mathbf{W}}_M^T, \hat{\mathbf{W}}_V^T, \hat{\mathbf{W}}_g^T]^T$ are the transformation and weight matrices of $\mathbf{M}$, $\mathbf{V}$, $\mathbf{g}$, respectively. The adaptive weight update laws are designed as

$$\dot{\hat{\mathbf{W}}}_M = -\iota_M (\sqsupset_M \dot{h}_J + \nu_M \hat{\mathbf{W}}_M) \\ \dot{\hat{\mathbf{W}}}_V = -\iota_V (\sqsupset_V \dot{h}_J + \nu_V \hat{\mathbf{W}}_V) \\ \dot{\hat{\mathbf{W}}}_g = -\iota_g (\sqsupset_g J + \nu_g \hat{\mathbf{W}}_g) \quad (16)$$

where ι_M , ι_V , and ι_g are the learning rate, ν_M , ν_V , and ν_g are the discontinuous switching constant defined as

$$\nu_\ell = \begin{cases} 0 & \text{if } \|\hat{\mathbf{W}}_\ell\| < \eta_\ell \\ \bar{\nu}_\ell & \text{if } \|\hat{\mathbf{W}}_\ell\| \geq \eta_\ell \end{cases}, \quad \ell = M, V, g \quad (17)$$

where η_ℓ is the weight updated threshold, and $\bar{v}_\ell$ is a manually designed positive constant. The term $v_\ell \hat{\mathbf{W}}_\ell$ in (16) is commonly applied to enhance the robustness of adaptive neural network controllers for dynamics [12]. However, it introduces pure damping to the weights during the learning process, potentially compromising approximation accuracy. To address this, (17) effectively mitigates this drawback by dynamically adjusting the weights based on the current value of $\|\hat{\mathbf{W}}_\ell\|$. Although the robot dynamic model exhibits inherent uncertainties, our controller can enable the robot to maintain adaptive and accurate control performance.

In fact, most robots are equipped with position/velocity control interfaces, and only a few come with joint torque or motor current interfaces. Therefore, based on the dynamic model (1), the joint angular acceleration can be derived as

$$\ddot{\mathbf{q}} = \mathbf{M}_q^{-1}(\mathbf{q})[\tau_c - \mathbf{J}^T \mathbf{f}_e - \mathbf{V}_q(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}_q(\mathbf{q})]. \quad (18)$$

Remark 4: The proposed ABBFNN controller incorporates bias BFNN and adaptive control gain adjustment, facilitating accurate estimation of the dynamic model. Furthermore, with the switching constant (17), the update laws dynamically adjust in real-time to prevent the weight from falling into local minima and to avoid oscillations.

C. Stability Analysis

Next, we perform Lyapunov stability analysis of the ABBFNN controller.

Theorem 1: For the dynamic system of the manipulator with control torque, the proposed ABBFNN controller can ensure that the tracking error is bounded and adaptively converge to guarantee uniformly ultimately bounded (UUB) stability.

Proof: Through (1), (13), and (15), and defining $(\hat{\cdot}) = (\cdot)^* - (\hat{\cdot})$, it can be derived as

$$\mathbf{M}_q^* \dot{\mathbf{j}} = \tilde{\mathbf{M}}_q \dot{\mathbf{h}} + \tilde{\mathbf{V}}_q \dot{\mathbf{h}} + \tilde{\mathbf{g}}_q - \mathbf{K}_{\dot{\mathbf{q}}} \mathbf{j} - \mathbf{e}_q - \mathbf{V}_q \mathbf{j}. \quad (19)$$

Design a Lyapunov function candidate as

$$\mathcal{V}_c = \frac{1}{2}(\mathbf{j}^T \mathbf{M}_q^* \mathbf{j} + \tilde{\mathbf{W}}_M^T \mathbf{M}_M^{-1} \tilde{\mathbf{W}}_M + \tilde{\mathbf{W}}_V^T \mathbf{V}_V^{-1} \tilde{\mathbf{W}}_V + \tilde{\mathbf{W}}_g^T \mathbf{g}_g^{-1} \tilde{\mathbf{W}}_g). \quad (20)$$

Substituting (16) and (19) into (20) and combining it with the skew-symmetric matrix $[\tilde{\mathbf{M}}_q - 2\mathbf{V}_q]$, it can be derived as

$$\begin{aligned} \dot{\mathcal{V}}_c &= -\mathbf{j}^T \mathbf{K}_{\dot{\mathbf{q}}} \mathbf{j} - \mathbf{j}^T \mathbf{e}_q \\ &\quad - \mathbf{v}_M \tilde{\mathbf{W}}_M^T \hat{\mathbf{W}}_M - \mathbf{v}_V \tilde{\mathbf{W}}_V^T \hat{\mathbf{W}}_V - \mathbf{v}_g \tilde{\mathbf{W}}_g^T \hat{\mathbf{W}}_g. \end{aligned}$$

According to Young's inequality $-\tilde{\mathbf{W}}^T \mathbf{W}^* \leq (1/2)\tilde{\mathbf{W}}^T \tilde{\mathbf{W}} + (1/2)\mathbf{W}^{*T} \mathbf{W}^*$ and $-\mathbf{j}^T \mathbf{e}_q \leq (1/2)\mathbf{j}^T \mathbf{j} + (1/2)\mathbf{e}_q^T \mathbf{e}_q$, it can be proved as

$$\begin{aligned} \dot{\mathcal{V}}_c &= -\mathbf{j}^T \mathbf{K}_{\dot{\mathbf{q}}} \mathbf{j} - \mathbf{j}^T \mathbf{e}_q \\ &\quad - \mathbf{v}_M \tilde{\mathbf{W}}_M^T \hat{\mathbf{W}}_M - \mathbf{v}_V \tilde{\mathbf{W}}_V^T \hat{\mathbf{W}}_V - \mathbf{v}_g \tilde{\mathbf{W}}_g^T \hat{\mathbf{W}}_g \\ &\leq -\mathbf{j}^T \mathbf{K}_{\dot{\mathbf{q}}} \mathbf{j} + \frac{1}{2}\mathbf{j}^T \mathbf{j} + \frac{1}{2}\mathbf{e}_q^T \mathbf{e}_q \\ &\quad - \frac{1}{2}(\mathbf{v}_M \tilde{\mathbf{W}}_M^T \tilde{\mathbf{W}}_M + \mathbf{v}_V \tilde{\mathbf{W}}_V^T \tilde{\mathbf{W}}_V + \mathbf{v}_g \tilde{\mathbf{W}}_g^T \tilde{\mathbf{W}}_g) \\ &\quad + \frac{1}{2}(\mathbf{v}_M \mathbf{W}_M^{*T} \mathbf{W}_M^* + \mathbf{v}_V \mathbf{W}_V^{*T} \mathbf{W}_V^* + \mathbf{v}_g \mathbf{W}_g^{*T} \mathbf{W}_g^*) \\ &\leq -\mathfrak{I} \mathcal{V}_c + \mathcal{T} \end{aligned} \quad (21)$$

where $\mathcal{T} = (1/2)(\mathbf{v}_M \mathbf{W}_M^{*T} \mathbf{W}_M^* + \mathbf{v}_V \mathbf{W}_V^{*T} \mathbf{W}_V^* + \mathbf{v}_g \mathbf{W}_g^{*T} \mathbf{W}_g^*) + (1/2)\mathbf{e}_q^T \mathbf{e}_q$, $\mathfrak{I} = \min\{(\mathbf{v}_M/\mathbf{t}_M^{-1}), (\mathbf{v}_V/\mathbf{t}_V^{-1}), (\mathbf{v}_g/\mathbf{t}_g^{-1}), (\lambda_{\min}(\mathbf{K}_{\dot{\mathbf{q}}}) - [1/2]/\lambda_{\max}(\mathbf{M}_q))\}$, in which $\lambda(\cdot)$ is the eigenvalue of $(\cdot)$. Then, we multiply (21) by $e^{\mathfrak{I}t}$ to get

$$\frac{d\mathcal{V}_c e^{\mathfrak{I}t}}{dt} \leq \mathcal{T} e^{\mathfrak{I}t}. \quad (22)$$

Next, we integrate (22) to obtain

$$\mathcal{V}_c \leq \mathcal{V}_c(0)e^{-\mathfrak{I}t} + \frac{\mathcal{T}}{\mathfrak{I}}(1 - e^{-\mathfrak{I}t}) \leq \mathcal{V}_c(0) + \frac{\mathcal{T}}{\mathfrak{I}}.$$

Since $\mathbf{M}_q$ is positive definite and bounded, the tracking error $\|\mathbf{e}_q\|$ and weight error $\|\tilde{\mathbf{W}}_\ell\|$ are bounded by $\|\mathbf{e}_q\| \leq \sqrt{2\mathcal{V}_c(0) + [\mathcal{T}/\mathfrak{I}]}$ and $\|\tilde{\mathbf{W}}_\ell\| \leq [\sqrt{2\mathcal{V}_c(0) + (\mathcal{T}/\mathfrak{I})}/\mathbf{t}_\ell^{-1}]$, $\ell = \mathbf{M}, \mathbf{V}, \mathbf{g}$, respectively. Therefore, it can be concluded that the tracking error $\mathbf{e}_q$ satisfies the requirement of UUB and the desired performance can be ensured by designing suitable parameters.

IV. ADAPTIVE OPTIMAL CONTROL FRAMEWORK DESIGN

In this section, we introduce the momentum-based force observer and construct the proposed adaptive optimal interaction control framework.

A. Momentum-Based Force Observer

Due to the potential interference and noise issues associated with existing sensors when measuring interaction force, as well as the increased hardware requirements, we introduced a sensorless momentum-based force observer used to accurately estimate the interaction force $\mathbf{f}_e$. According to (1) and $\dot{\mathbf{M}}_q = \mathbf{V}_q + \mathbf{V}_q^T$, the derivative of the momentum $\mathbf{p} = \mathbf{M}_q(\mathbf{q})\dot{\mathbf{q}}$ can be expressed as

$$\begin{aligned} \dot{\mathbf{p}} &= \dot{\mathbf{M}}_q(\mathbf{q})\dot{\mathbf{q}} - \mathbf{V}_q(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}_q(\mathbf{q}) + \tau_c - \tau_e \\ &= \mathbf{V}_q^T(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}_q(\mathbf{q}) + \tau_c - \tau_e. \end{aligned}$$

Furthermore, a residual term $\boldsymbol{\gamma} \in \mathcal{R}^m$ is introduced to estimate the interaction torque τ_e as

$$\boldsymbol{\gamma} = \mathbf{G}_o \left(\hat{\mathbf{p}}(t) - \int_0^t (\hat{\boldsymbol{\Theta}}_q(\mathbf{q}, \dot{\mathbf{q}}) + \tau_c - \boldsymbol{\gamma}) dt - \mathbf{p}(0) \right)$$

where $\mathbf{G}_o = \text{diag}\{g_{o,1}, g_{o,2}, \dots, g_{o,n}\} > 0 \in \mathcal{R}^{n \times n}$ is the gain matrix, $\hat{\mathbf{p}} = \hat{\mathbf{M}}_q(\mathbf{q})\dot{\mathbf{q}}$, $\hat{\boldsymbol{\Theta}}_q(\mathbf{q}, \dot{\mathbf{q}}) = \hat{\mathbf{V}}_q^T(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \hat{\mathbf{g}}_q(\mathbf{q})$, in which $(\hat{\cdot})$ represents the estimation result.

Next, the dynamics of the momentum-based force observer is defined as

$$\begin{aligned} \dot{\hat{\mathbf{p}}} &= \hat{\boldsymbol{\Theta}}_q(\mathbf{q}, \dot{\mathbf{q}}) + \tau_c - \boldsymbol{\gamma} \\ \dot{\boldsymbol{\gamma}} &= \mathbf{G}_o(\dot{\hat{\mathbf{p}}} - \dot{\hat{\mathbf{p}}}). \end{aligned}$$

Under the assumption that the robot dynamic models are optimal, $\dot{\boldsymbol{\gamma}}$ can be expressed as

$$\dot{\boldsymbol{\gamma}} = \mathbf{G}_o(\tau_e - \boldsymbol{\gamma}). \quad (23)$$

By applying the Laplace transform, (23) can be rewritten as

$$\frac{\gamma_i}{\tau_{e,i}} = \frac{g_i}{s + g_i} = \frac{1}{1 + s/g_i}, \quad i = 1, \dots, n$$

where γ_i is the i th term of $\boldsymbol{\gamma}$. It is obvious that as $g_i \rightarrow +\infty$, $\gamma_i \rightarrow \tau_{e,i}$, which implies the virtual force sensor capable

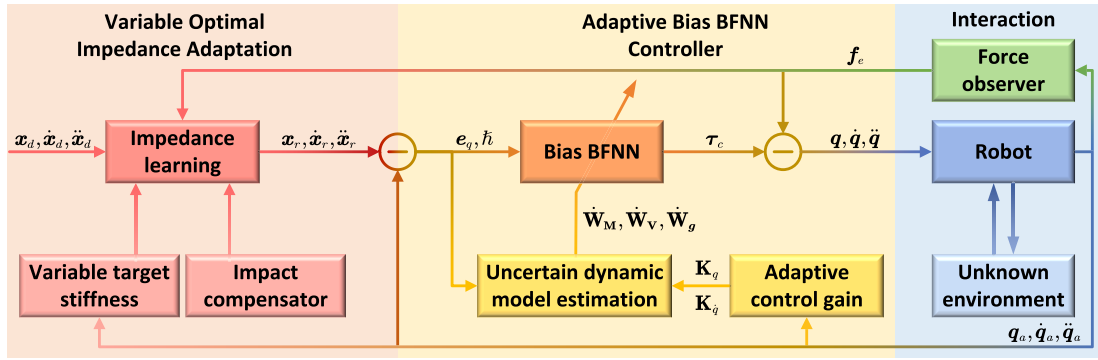


Fig. 4. Structure of adaptive optimal interaction control framework.

of converting momentum observations into external joint torques. Further, torque can be transformed into force using the Jacobian matrix as $\hat{f}_e = (J^T)^\dagger \tau$, where $(\cdot)^\dagger$ represents the pseudo-inverse operation. Finally, to mitigate the effects of noise, a Kalman filter [40] is employed for smoothing.

Remark 5: The interaction force f_e between the robot and the environment is estimated in real-time by the momentum-based force observer and fed back to the VOIA and ABBFNN controller. This method enhances data accuracy and learning efficiency through force estimation while mitigating signal interference. In contrast to directly attaching a force sensor to the end effector, it effectively reduces the complexity of the control system and hardware requirements.

B. Overview of the Framework

The framework structure is illustrated in Fig. 4. In this framework, VOIA generates a reference trajectory x_r by optimizing the variable impedance model (6), aiming to achieve optimal interaction with minimal position and force errors. Throughout the learning process, the environment model (2) integrates variable target stiffness K_v (4) and impact compensator B_v (5) for real-time adjustments. First, VOIA combines exploration noise ϵ to calculate the operators (7). Next, it iteratively solves (8) to obtain the optimal impedance gain G^* . Then, a soft function (9) is introduced to dynamically smooth G . Finally, it calculates the reference trajectory x_r using the approximate optimal interaction force.

Once x_r is obtained, it is fed into the ABBFNN controller without the need for training. With real-time feedback incorporating motion and force information, the controller adaptively adjusts the control gains (14). First, ABBFNN calculates the joint space reference trajectory $q_r, \dot{q}_r, \ddot{q}_r$ and the virtual control term $\tilde{h}$. Then, it determines whether to add network nodes centered at (11) by (10). Next, it updates the network parameters (12) and approximates the uncertain robot dynamic model by (15). Finally, the joint control torque τ_c (13) is yielded based on the approximation results, which is then transformed into a more universally applicable joint motion control signal by (18).

In conclusion, the proposed adaptive optimal interaction control framework emulates human arm muscle characteristics, achieving optimal interaction with minimal combined position

and force errors. By adaptively optimizing reference trajectories through VOIA and leveraging the ABBFNN controller's accurate tracking and parameter adaptation capability, this framework contributes to realize safe and friendly interactions, facilitating human-like operations.

Remark 6: While the adaptive parameter adjustment in the framework can adapt to environment changes and accelerate error convergence, it may lead to an increase in computational overhead. Therefore, hardware acceleration techniques can be explored, and the algorithm can be further optimized by incremental dynamic adjustment of the update step.

V. SIMULATIONS AND REAL-WORLD EXPERIMENTS

In this section, we validate the proposed method through simulations, real-world experiments, and comparative analyses and employ the UR5 manipulator as experimental platform.

A. Verification of Variable Optimal Impedance Adaptation

In this part, we use VOIA to obtain complex reference trajectories of the manipulator interacting with an unknown plane whose z-axis is at 0.15 m. Without loss of generality, we focus on the force in z-direction while disregarding friction.

Furthermore, the parameter settings in VOIA are as follows: The initial position of the end-effector is set as $[-0.5, -0.1, 0.16]$ m. The running time T is set as 5 s. The parameters of the intermediate state variable (3) are set as $A = -1$ and $E = 0.1$. The proportional, integral, and differential gains are set as $k_p = 0.4$, $k_i = 0.08$, and $k_d = 0.01$. The impact evaluation parameters are set as $\eta_1 = 0.1$, $\eta_2 = 5$. The tracking error weight N is set as 100, and the interaction force weight H is set as 1. The exploration noise ϵ is set as $\epsilon = \sum_{n_\epsilon=1}^{10} (0.02/n_\epsilon) \sin(n_\epsilon t)$. The state of the environment is assumed as $0.01\dot{x} + (x - x_d) = -f_e$. The initial impedance gain is set as $g_0 = [-0.1, 0.1]$. The desired trajectories are two complex curves: butterfly curve and rose curve, defined as

$$x_{d,1} = \begin{bmatrix} 0.035 \cos(2\pi t)(\exp(\cos(2\pi t)) - 2 \cos(8\pi t) - \sin(\pi t/12)^5) - 0.5 - 0.035(\exp(1) - 2) \\ 0.035 \sin(2\pi t)(\exp(\cos(2\pi t)) - 2 \cos(8\pi t) - \sin(\pi t/12)^5) - 0.1 \\ 0.125 \end{bmatrix}$$

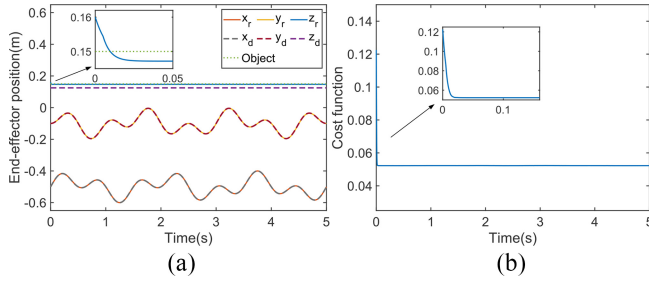


Fig. 5. Interaction tracking results of VOIA. (a) Tracking results of rose curve. (b) Cost function value.

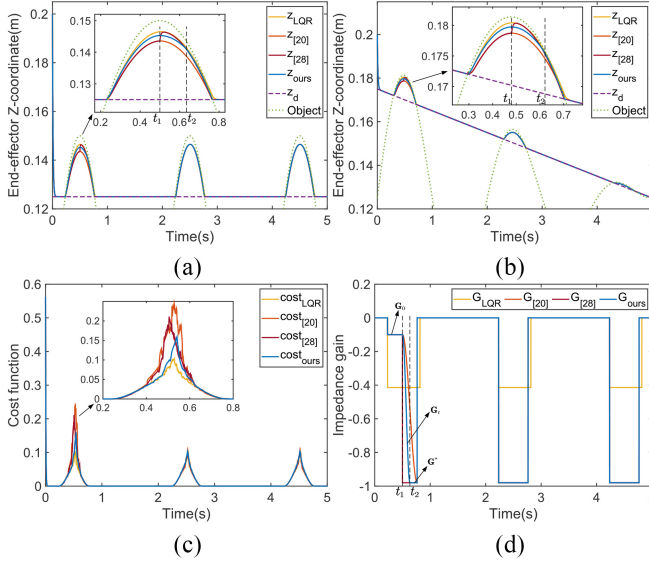


Fig. 6. Comparison of VOIA with LQR, [20], [28]. (a) and (b) Z-axis interaction tracking results. (c) Cost function value. (d) Evolution of the impedance gain $\mathbf{G}$.

$$\mathbf{x}_{d,2} = \begin{bmatrix} 0.1 \cos(4\pi t) \sin(10\pi t) \\ 0.1 \sin(4\pi t) \sin(10\pi t) \\ 0.125 \end{bmatrix}.$$

The generated reference trajectory from the rose curve and the variation of the cost function are depicted in Fig. 5. It can be observed that the proposed method can yield a smooth reference trajectory. In terms of the cost function, it progressively converges with the learning of the optimization feedback gain $\mathbf{G}^*$. Subsequently, to demonstrate the benefits of VOIA, we consider complex surface objects as the interaction target and conduct a comparative analysis with LQR, impedance adaptation [28], and method in [20]. The two complex surfaces are defined as

$$\begin{aligned} z_{o,1} &= \max\{0.125, 0.075 \sin(10\pi(y + 0.25)) + 0.075\} \\ z_{o,2} &= \max\{-0.1y + 0.15 \\ &\quad 0.075 \sin(10\pi(y + 0.25))(-2y + 0.5) + 0.125\}. \end{aligned}$$

Moreover, the desired trajectories to be tracked are defined as $\mathbf{x}_{d,1} = [-0.5, 0.1t - 0.25, 0.125]$ and $\mathbf{x}_{d,2} = [-0.5, 0.1t - 0.25, -0.1y + 0.15]$. The initial position of the end-effector is set as $[-0.5, -0.25, 0.2]$ m.

Fig. 6 illustrates the interaction results of the above methods. It is evident that by incorporating the novel variable

TABLE II
QUANTITATIVE COMPARISON OF VOIA WITH LQR, [20], [28] DURING IMPEDANCE LEARNING

Method	e_{avg} (m)	f_{avg} (N)	cost_{avg}	cost_{max}
LQR	0.0142	0.0639	0.0323	0.1030
[20]	0.0120	0.1434	0.0557	0.2454
[28]	0.0130	0.1246	0.0490	0.2112
Ours	0.0135	0.0938	0.0385	0.1596

target stiffness and impact compensator, the proposed method can achieve a closer effect to the optimal baseline LQR, which interacts with a priori known environment. As shown in Fig. 6(c) and (d), VOIA can achieve a lower cost function than [20], [28], effectively optimizing the interaction trajectory during the learning phase with sudden impact. Additionally, the trajectory becomes smoother when combined with the auxiliary soft function $\mathbf{G}_t$. Hence, it is demonstrated that the proposed VOIA method can generate smooth and continuous reference trajectories when facing complex interaction tasks. Moreover, this method can achieve robust interactions in unknown environments, making its application scope broader than LQR. Furthermore, Table II provides a quantitative comparison of the average position error e_{avg} , average force error f_{avg} , average cost function value cost_{avg} , and maximum cost function value cost_{max} among the four methods. It is evident that our method more closely approximates LQR, which requires prior knowledge of the environment, compared to [28] and [20] for unknown environment models. The average cost function value of our method is only 19.19% higher than that of LQR, whereas the other two methods show increases of 72.44% and 51.70%, respectively. This demonstrates that VOIA achieves superior interaction performance in unknown environments, enabling safe operation.

B. Verification of Adaptive Bias BFNN Controller

In this part, we utilize the ABBFNN controller to track the complex trajectories mentioned above and interact with the plane. Furthermore, we conduct an ablation study and compare with bias RBFNN [12] and BFNN [20]. The parameter settings in ABBFNN are as follows.

The number of closest centers $n_{\min}$ is chosen as 3. The distance threshold is set as $\mathbf{t} = [0.2, 0.3, 0.25]$. The initial bias is set as $\mathbf{b} = [0.5, 0.5, 0.5, 0.5, 0.2, 0.2]^T$. The weight adjustment threshold is set as $\chi = [0.002, 0.002, 0.002, 0.002, 0.05, 0.05]$. The gain adjustment step ρ is set as 5. The learning rates are set as $\mathbf{t}_{\mathbf{M}} = \mathbf{t}_{\mathbf{V}} = \mathbf{t}_{\mathbf{g}} = 1.5$, $\bar{\mathbf{v}}_{\mathbf{M}} = \bar{\mathbf{v}}_{\mathbf{V}} = \bar{\mathbf{v}}_{\mathbf{g}} = 5$. The weight updated thresholds are set as $\eta_{\ell} = 1$.

Fig. 7 illustrates the motion of the manipulator tracking these two complex trajectories. For a more in-depth analysis of its performance, Fig. 8 presents the control results for the rose curve. It is evident from the changes in joint angles and end-effector position that the manipulator operates smoothly. Furthermore, the tracking error gradually converges and stabilizes at the level of 10^{-5} m within 0.1 s. Additionally, Fig. 9 shows the approximate uncertain dynamics, and the deviations between the approximate uncertain dynamics and the assumed values of the actual uncertain dynamics, respectively. Table III

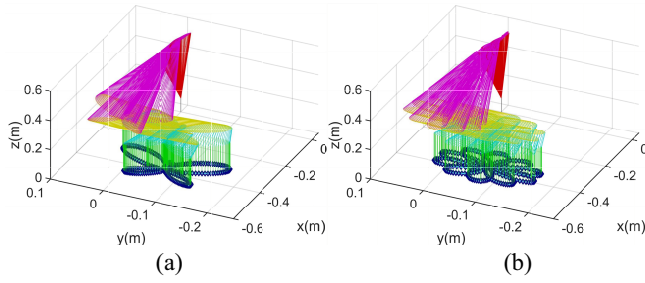


Fig. 7. Motion of the manipulator on different curves. (a) Butterfly curve. (b) Rose curve.

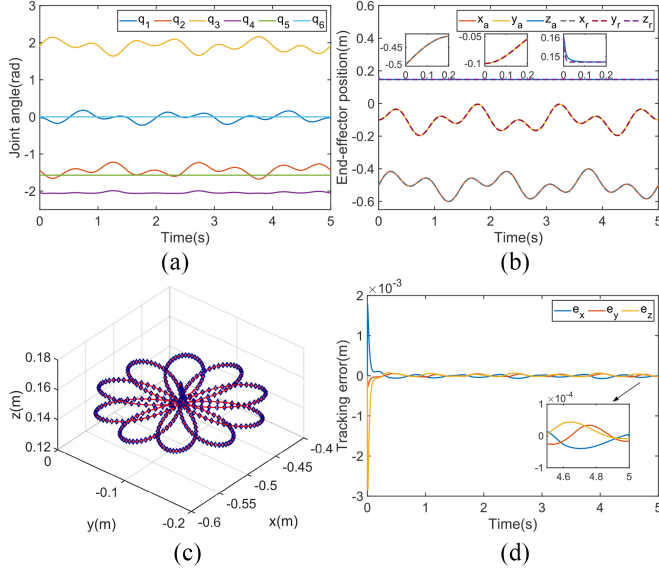


Fig. 8. Control results of ABBFNN. (a) Joint angle of the manipulator. (b) End-effector position of the manipulator. (c) Desired and actual trajectory of the end-effector. (d) Trajectory tracking error.

TABLE III
CONVERGENCE TIME AND POST-CONVERGENCE ERROR OF THE
APPROXIMATE UNCERTAIN DYNAMICS

Dynamics	t_{conv} (s)	e_{avg}	e_{max}
$\mathbf{M}_q$	0.127	1.07×10^{-5}	1.81×10^{-4}
$\mathbf{V}_q$	0.097	6.26×10^{-4}	5.86×10^{-3}
$\mathbf{g}_q$	0.245	3.49×10^{-4}	1.41×10^{-3}

presents the convergence time, average and maximum convergence error. It can be observed that the adaptive weight update laws (16) can actively adapt to uncertain dynamics, achieving convergence within 0.25 s with an average approximation error stabilized at the level of 10^{-3} . This demonstrates that the ABBFNN controller can mitigate deviations in robot parameters arising from measurement errors, mechanical wear and interactions, thereby achieving an accurate approximation of uncertain dynamic model and improving control performance.

The changes in center points and enhancement nodes while estimating dynamic parameters are depicted in Fig. 10. As the manipulator moves, the centers gradually spread across the entire trajectory, and the number of enhancement nodes increases correspondingly, facilitating the estimation of uncertain dynamic model. Fig. 10(d) shows the number of nodes

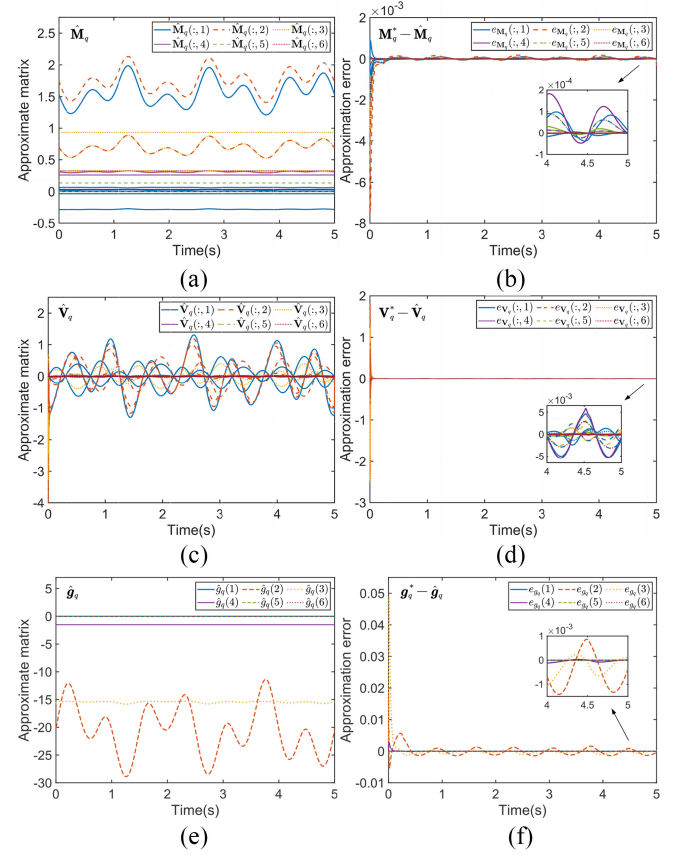


Fig. 9. Approximate uncertain dynamics and the deviation from actual dynamics. (a) and (b) Inertia matrix $\mathbf{M}_q$. (c) and (d) Coriolis and centrifugal matrix $\mathbf{V}_q$. (e) and (f) Gravity vector $\mathbf{g}_q$.

obtained in different ι , where $\iota_1 = [0.175, 0.275, 0.225]$, $\iota_2 = [0.2, 0.3, 0.25]$, $\iota_3 = [0.24, 0.32, 0.28]$. It can be seen that the number of enhancement nodes adjusts adaptively with the change in the distance threshold, which effectively reduces the number of neural network parameters and design requirements.

Next, to demonstrate the effectiveness of bias and adaptive gain adjustment, an ablation study is conducted, and the tracking errors for bias BFNN (BBFNN) and adaptive BFNN (ABFNN) are shown in Fig. 11. The dashed line represents the baseline results, i.e., the tracking error without improvements. Clearly, compared to the baseline, the bias effectively mitigates post-convergence oscillations, reducing the tracking error. Meanwhile, adaptive gain adjustment reduces the overall tracking error throughout the process, effectively accelerates the rate of convergence, and minimizes the convergence time.

Then, comparison experiments are conducted. The control results of bias RBFNN [12] and BFNN [20] are shown in Fig. 12. The dashed line represents the results of the proposed method. It can be observed that the tracking error of adaptive bias RBFNN is large, reaching 0.015 m in the initial interaction stage, making it ineffective to perform complex trajectory tracking tasks effectively. BFNN demonstrates good control performance but exhibits slow convergence and large error fluctuation with oscillation phenomena.

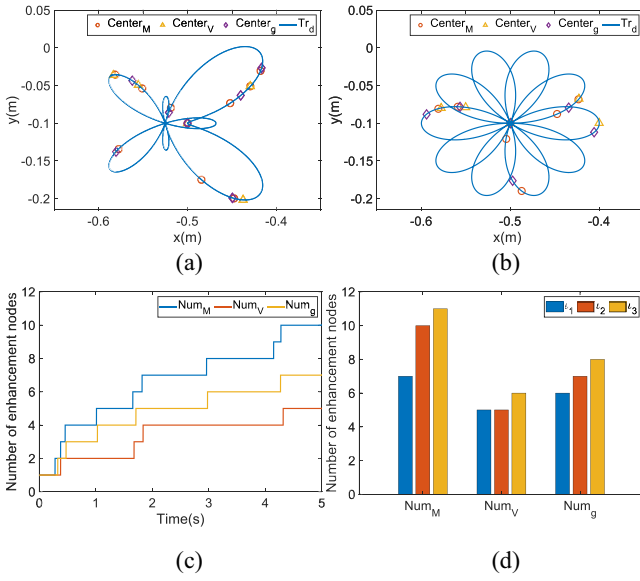


Fig. 10. Change of centers μ and enhancement nodes Ψ . (a) and (b) Centers of the end-effector. (c) and (d) Number of enhancement nodes.

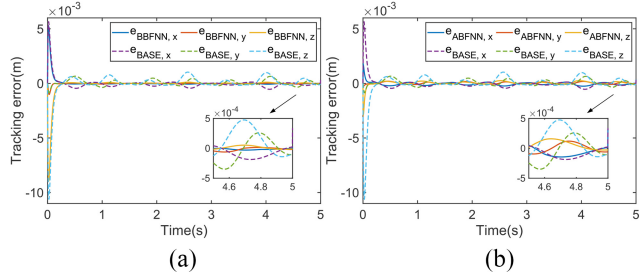


Fig. 11. Ablation study. (a) Tracking error of BBFNN. (b) Tracking error of ABFNN.

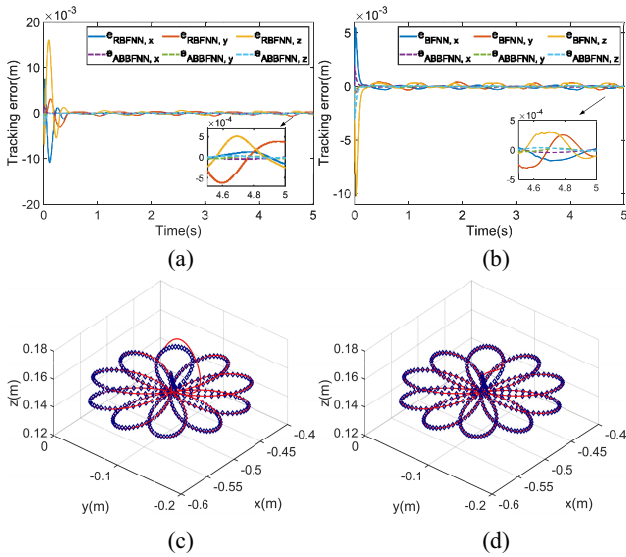


Fig. 12. Comparison of ABFNN with other methods. (a) and (b) Tracking error of bias RBFNN [12] and BFNN [20]. (c) and (d) Desired and actual trajectory obtained by [12] and [20].

Finally, Table IV presents the convergence time, average and maximum convergence error of the ablation study and comparison experiments. It is obvious that bias RBFNN [12]

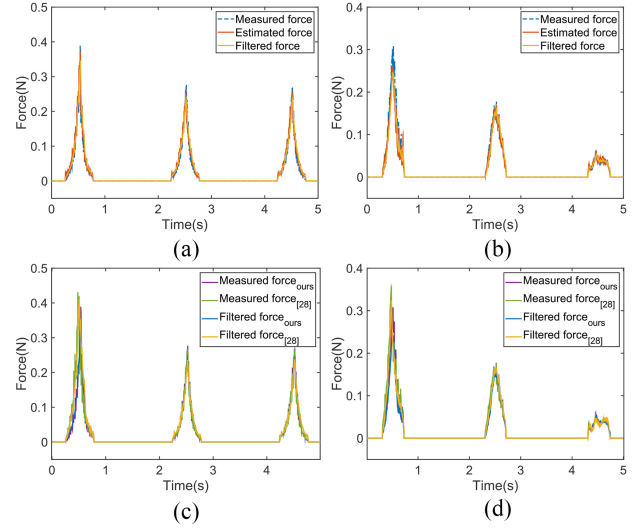


Fig. 13. Contact force in interaction control tasks. (a) and (b) Measured and estimated forces for interaction with complex surfaces. (c) and (d) Interaction forces of the proposed method and [28].

TABLE IV
QUANTITATIVE ABLATION STUDY AND COMPARISON OF ABFNN WITH [12] AND [20]

Method	t_{conv} (s)	e_{avg} (m)	e_{max} (m)
Bias RBFNN [12]	0.489	4.88×10^{-4}	6.20×10^{-4}
BFNN [20]	0.293	2.13×10^{-4}	3.68×10^{-4}
BBFNN	0.278	8.57×10^{-5}	1.09×10^{-4}
ABFNN	0.112	8.64×10^{-5}	3.30×10^{-4}
ABFNN	0.091	3.25×10^{-5}	7.01×10^{-5}

TABLE V
QUANTITATIVE RESULTS OF CONTACT FORCE IN INTERACTION TASKS

Task	$e_{f,avg}$ (N)	$e_{f,max}$ (N)
Task 1 (Fig. 13 (a))	0.0033	0.0653
Task 2 (Fig. 13 (b))	0.0034	0.0743
Average	0.0034	0.0698

performs modestly in terms of convergence time and error, with the longest convergence time of 0.489 s. While BFNN [20] improves these metrics to some extent, its high post-convergence oscillations lead to an elevated average convergence error at 2.13×10^{-4} m. In contrast, our adaptive gain adjustment effectively reduces convergence time to 0.112 s, and the bias significantly enhances post-convergence stability, with a maximum error of only 1.09×10^{-4} m. Consequently, ABFNN combines these advantages, achieving the fastest convergence at 0.091 s with the smallest convergence error of 3.25×10^{-5} m and 7.01×10^{-5} m. On the whole, ABFNN demonstrates excellent tracking control performance, in which the incorporation of bias and adaptive parameters effectively reduces tracking errors and improves the rate of convergence.

C. Verification of the Momentum-Based Force Observer

In this part, we validate the estimation performance of the momentum-based force observer and demonstrate the effectiveness of the impact compensator.

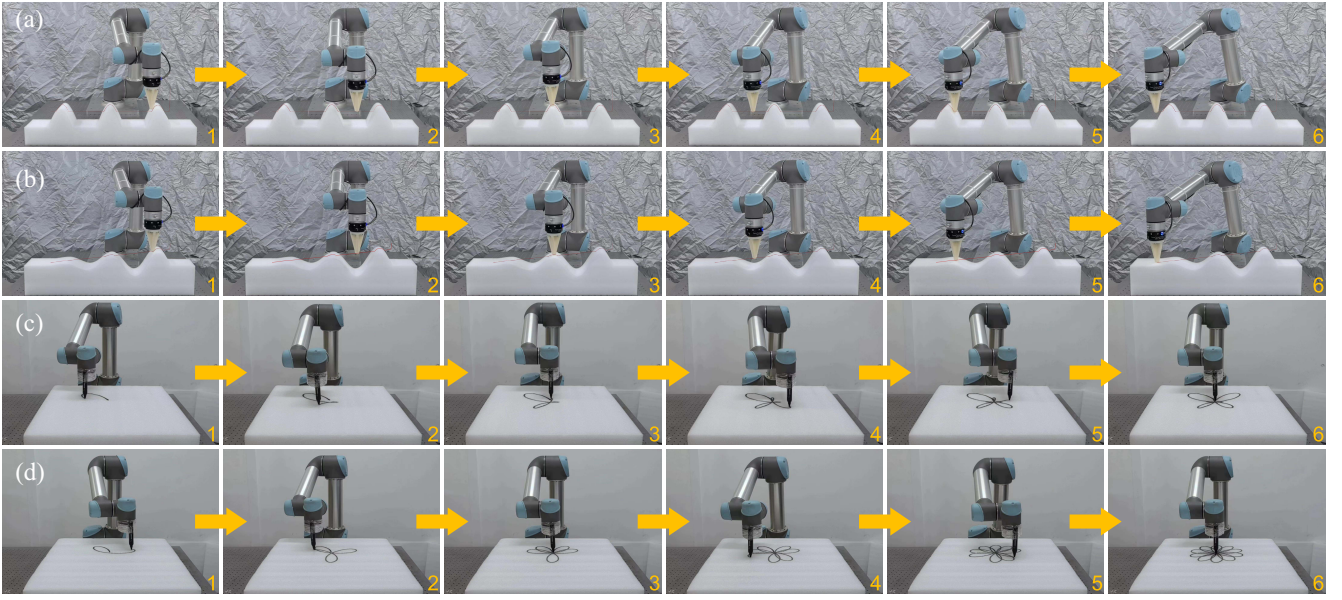


Fig. 14. Experiments of different manipulator interaction control tasks.

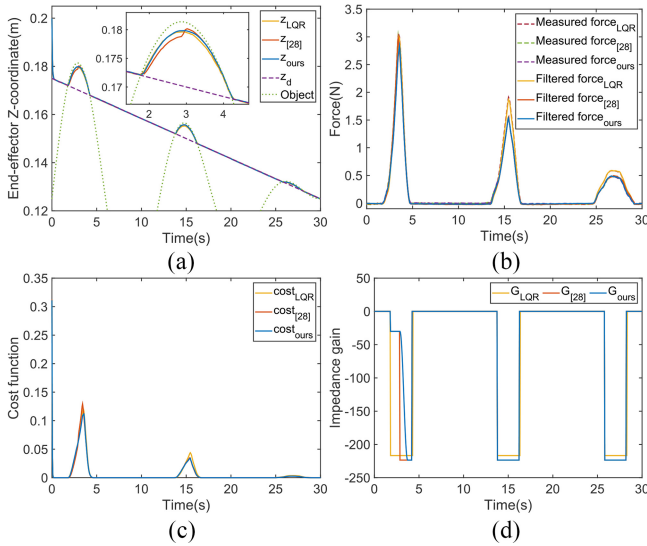


Fig. 15. Interaction results and comparison of the proposed framework. (a) Z-axis interaction tracking results. (b) Measured and estimated interaction forces. (c) Value of cost function. (d) Evolution of the impedance gain $\mathbf{G}$.

TABLE VI
QUANTITATIVE COMPARISON OF THE PROPOSED FRAMEWORK IN
REAL-WORLD EXPERIMENT

Method	e_{avg} (m)	f_{avg} (N)	$e_{f,\text{avg}}$ (N)	cost_{avg}
LQR	0.0216	1.2884	0.0118	0.0499
[28]	0.0210	1.2979	0.0148	0.0488
Ours	0.0213	1.1120	0.0118	0.0476

Fig. 13 illustrates the momentum-based force observer's evaluation during interaction, with the corresponding quantitative results shown in Table V. It can be seen that the observer excels with minimal error, with an average force estimation error of 0.003 N in both tasks and a maximum error of less than 0.075 N, ensuring accurate tracking of external force variations during tasks. Furthermore, with the addition of impact compensator, forces during sudden interactions are

reduced, and vibration during the operation is effectively suppressed. Hence, it validates the force estimation capability of the momentum-based force observer and demonstrates the impact damper's ability to reduce instantaneous impact forces.

D. Implementation and Verification

In this part, we conduct real-world experiments, where the manipulator is equipped with a Robotiq FT300 Force/Torque Sensor. The computer communicates with the robot controller and the force/moment transducer via TCP/IP protocol. As to the parameters, the initial impedance gain is set as $\mathbf{G}_0 = [-30, 1]$. The tracking error weight $\mathbf{N}$ is set as 500, and the interaction force weight $\mathbf{H}$ is set as 0.01. The state of the environment is assumed as $0.4\dot{\mathbf{x}} + 7(\mathbf{x} - \mathbf{x}_d) = -\mathbf{f}_e$. The other parameters use the previous definitions. Fig. 14 shows the interaction control process, in which the interaction objects are composed of expandable polyethylene (EPE), further validating the proposed framework's practicality and effectiveness.

The complex surface interaction results of Fig. 14(b) are shown in Fig. 15, and compared with LQR and [28]. And the corresponding quantized data is shown in Table VI. It is evident that the performance of LQR declines significantly due to the unknown model of EPE. In contrast, our framework still maintains good interaction capability. Furthermore, despite the presence of interaction and noninteraction stages due to changes in the interaction objects, the proposed framework demonstrates strong adaptability to such changes. It can autonomously adjust parameters, mitigate impacts, and quickly learn impedance gain to achieve optimal interaction performance during interaction. Moreover, the force observer exhibits good estimation capability, thus effectively simplifying the control system.

The trajectory tracking results of Fig. 14(c) are illustrated in Fig. 16. It can be seen that the proposed control framework demonstrates effective tracking performance, consistent with

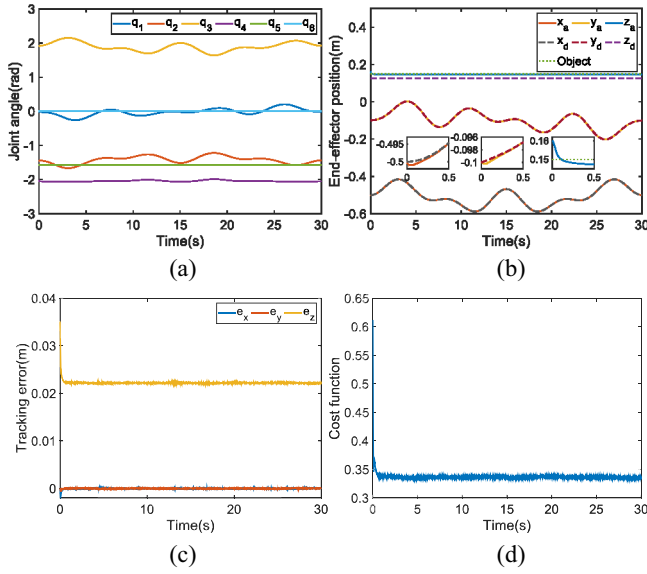


Fig. 16. Tracking results of the proposed framework. (a) Joint angle changes in tracking control. (b) End-effector position of the manipulator. (c) Trajectory tracking error. (d) Value of cost function.

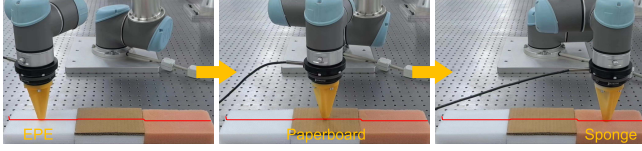


Fig. 17. Experiment when the material of environment changes.

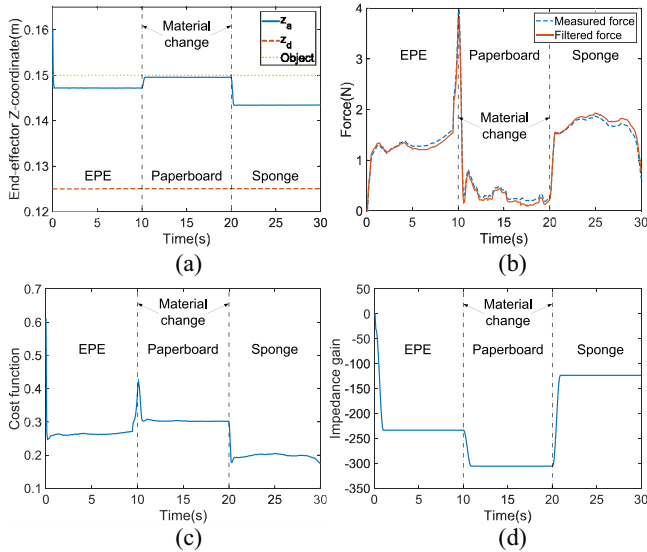


Fig. 18. Interaction results when the material of the environment changes. (a) Z-axis interaction tracking results. (b) Measured and estimated interaction forces. (c) Value of cost function. (d) Evolution of the impedance gain G .

the simulation results. The tracking error is able to converge within 0.5 s. Due to the presence of interaction objects, the error between z_a and z_d stabilizes at 0.022 m, thereby minimizing the cost function (6) composed of tracking error and force error.

Moreover, to further validate the framework's adaptability to changes in the environment, we conducted an experiment

where the material of the environment is altered. The process and results are shown in Figs. 17 and 18. It can be observed that despite the sudden changes in interaction forces due to material variations, the proposed framework effectively adapts to these changes by learning the impedance gain and adjusting the trajectory to achieve friendly interaction with different materials. The evolution of cost function further demonstrates the framework's adaptability and robustness to changes.

Overall, employing our framework for robot interacting with unknown environment enables efficient, convenient, and accurate control, proving effective in minimizing the requirements of environment knowledge and parameter design.

VI. CONCLUSION

This article proposes a novel adaptive optimization interaction control framework, inspired by the characteristics of human arm muscles, for optimal interaction between uncertain dynamic robots and unknown environments. The framework integrates VOIA and ABBFNN controller to ensure that the robot can achieve smooth operations and adapt to changes during interactions, demonstrating good adaptation, model approximation, optimal interaction, powerful tracking, and oscillation suppression capabilities. It is theoretically proved that the controller is UUB stable under the constraint Lyapunov function. Simulation, experimentation, and comparisons validate the effectiveness of the framework in dealing with changing interaction tasks, showing its potential to improve safety, promote human-like operations, and facilitate the practical application of robots in various fields. It is of great practical significance for further addressing robot-environment interactions in complex scenarios. Future work will focus on exploring ways to improve the framework's performance, particularly in complex environments with disturbances, by introducing more efficient optimization algorithms and leveraging interactive behavior prediction models.

REFERENCES

- [1] Y. Xie et al., "Robot-assisted transesophageal ultrasound and the virtual admittance-based master-slave control method thereof," *IEEE/ASME Trans. Mechatronics*, vol. 28, no. 5, pp. 2505–2516, Oct. 2023.
- [2] D. Guo, R. Y. Zhong, Y. Rong, and G. G. Q. Huang, "Synchronization of shop-floor logistics and manufacturing under IIoT and digital twin-enabled graduation intelligent manufacturing system," *IEEE Trans. Cybern.*, vol. 53, no. 3, pp. 2005–2016, Mar. 2023.
- [3] Z. Li, C. Xu, Q. Wei, C. Shi, and C.-Y. Su, "Human-inspired control of dual-arm exoskeleton robots with force and impedance adaptation," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 12, pp. 5296–5305, Dec. 2020.
- [4] C. Yang, G. Peng, Y. Li, R. Cui, L. Cheng, and Z. Li, "Neural networks enhanced adaptive admittance control of optimized robot-environment interaction," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2568–2579, Jul. 2019.
- [5] S. H. Yan, D. Xu, and X. Tao, "Hierarchical policy learning with demonstration learning for robotic multiple peg-in-hole assembly tasks," *IEEE Trans. Ind. Informat.*, vol. 19, no. 10, pp. 10254–10264, Oct. 2023.
- [6] H. Huang et al., "Robust passivity-based dynamical systems for compliant motion adaptation," *IEEE/ASME Trans. Mechatronics*, vol. 27, no. 6, pp. 4819–4828, Dec. 2022.
- [7] G. Peng, C. L. P. Chen, and C. Yang, "Robust admittance control of optimized robot-environment interaction using reference adaptation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 9, pp. 5804–5815, Sep. 2023.

- [8] X. Li, Q. Lu, J. Chen, N. Jiang, and K. Li, "A robust region control approach for simultaneous trajectory tracking and compliant physical human-robot interaction," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 10, pp. 6388–6400, Oct. 2023.
- [9] T. Poggio and F. Girosi, "Networks for approximation and learning," *Proc. IEEE*, vol. 78, no. 9, pp. 1481–1497, Sep. 1990.
- [10] C. Yang, D. Huang, W. He, and L. Cheng, "Neural control of robot manipulators with trajectory tracking constraints and input saturation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 9, pp. 4231–4242, Sep. 2021.
- [11] G. Peng, C. L. P. Chen, and C. Yang, "Neural networks enhanced optimal admittance control of robot-environment interaction using reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4551–4561, Sep. 2022.
- [12] Q. Liu, D. Li, S. S. Ge, R. Ji, Z. Ouyang, and K. P. Tee, "Adaptive bias rbf neural network control for a robotic manipulator," *Neurocomput.*, vol. 447, pp. 213–223, Aug. 2021.
- [13] H. O. Wang, K. Tanaka, and M. F. Griffin, "An approach to fuzzy control of nonlinear systems: Stability and design issues," *IEEE Trans. Fuzzy Syst.*, vol. 4, no. 1, pp. 14–23, Feb. 1996.
- [14] C. Wu, J. Liu, X. Jing, H. Li, and L. Wu, "Adaptive fuzzy control for nonlinear networked control systems," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 47, no. 8, pp. 2420–2430, Aug. 2017.
- [15] Q. Wang, J. Cao, and H. Liu, "Adaptive fuzzy control of nonlinear systems with predefined time and accuracy," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 12, pp. 5152–5165, Dec. 2022.
- [16] W. He and Y. Dong, "Adaptive fuzzy neural network control for a constrained robot using impedance learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 4, pp. 1174–1186, Apr. 2018.
- [17] H. Ma, Q. Zhou, H. Li, and R. Lu, "Adaptive prescribed performance control of a flexible-joint robotic manipulator with dynamic uncertainties," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 12905–12915, Dec. 2022.
- [18] G. Li, J. Yu, and X. Chen, "Adaptive fuzzy neural network command filtered impedance control of constrained robotic manipulators with disturbance observer," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 5171–5180, Aug. 2023.
- [19] H. Huang, C. Yang, Z. Ju, Y. Yuan, and Z. Li, "Broad fuzzy neural control using impedance learning," in *Proc. IEEE 4th Int. Conf. Adv. Robot. Mechatronics (ICARM)*, Toyonaka, Japan, 2019, pp. 173–178.
- [20] H. Huang, C. Yang, and C. L. P. Chen, "Optimal robot-environment interaction under broad fuzzy neural adaptive control," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3824–3835, Jul. 2021.
- [21] C. L. P. Chen and Z. Liu, "Broad learning system: An effective and efficient incremental learning system without the need for deep architecture," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 1, pp. 10–24, Jan. 2018.
- [22] C. P. Cop, K. L. Jakubowski, A. C. Schouten, B. Koopman, E. J. Perreault, and M. Sartori, "The simultaneous model-based estimation of joint, muscle, and tendon stiffness is highly sensitive to the tendon force-strain relationship," *IEEE Trans. Bio-Med. Eng.*, vol. 71, no. 3, pp. 987–997, Mar. 2024.
- [23] L. Yao, B. Zhao, X. Xu, Z. Wang, P. K. Wong, and Y. Hu, "Efficient incremental offline reinforcement learning with sparse broad critic approximation," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 1, pp. 156–169, Jan. 2024.
- [24] E. Arefinia, H. A. Talebi, and A. Doustmohammadi, "A robust adaptive model reference impedance control of a robotic manipulator with actuator saturation," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 2, pp. 409–420, Feb. 2020.
- [25] R. Johansson and M. Spong, "Quadratic optimization of impedance control," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, San Diego, CA, USA, 1994, pp. 616–621.
- [26] M. Matinfar and K. Hashtrudi-Zaad, "Optimization-based robot compliance control: geometric and linear quadratic approaches," *Int. J. Robot. Res.*, vol. 24, no. 8, pp. 645–656, 2005.
- [27] S. Hussain, S. Q. Xie, and P. K. Jamwal, "Adaptive impedance control of a robotic orthosis for gait rehabilitation," *IEEE Trans. Cybern.*, vol. 43, no. 3, pp. 1025–1034, Jun. 2013.
- [28] S. S. Ge, Y. Li, and C. Wang, "Impedance adaptation for optimal robot-environment interaction," *Int. J. Control*, vol. 87, no. 2, pp. 249–263, 2014.
- [29] Y. Jiang and Z.-P. Jiang, "Computational adaptive optimal control for continuous-time linear systems with completely unknown dynamics," *Automatica*, vol. 48, no. 10, pp. 2699–2704, 2012.
- [30] L. Roveda, A. A. Shahid, N. Iannacci, and D. Piga, "Sensorless optimal interaction control exploiting environment stiffness estimation," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 1, pp. 218–233, Jan. 2022.
- [31] Y. Zhang, X. Zhao, Y. Chen, B. Tao, and H. Ding, "Observer-based variable impedance control using moving horizon estimation for robot machining thin-walled workpieces," *IEEE Trans. Ind. Electron.*, vol. 71, no. 6, pp. 5972–5982, Jun. 2024.
- [32] X. Xing, E. Burdet, W. Si, C. Yang, and Y. Li, "Impedance learning for human-guided robots in contact with unknown environments," *IEEE Trans. Robot.*, vol. 39, no. 5, pp. 3705–3721, Oct. 2023.
- [33] C. Yang, G. Peng, L. Cheng, J. Na, and Z. Li, "Force sensorless admittance control for teleoperation of uncertain robot manipulator using neural networks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 51, no. 5, pp. 3282–3292, May 2021.
- [34] X. Zhang, H. Zhou, J. Liu, Z. Ju, Y. Leng, and C. Yang, "A practical PID variable stiffness control and its enhancement for compliant force-tracking interactions with unknown environments," *Sci. China Technol. Sci.*, vol. 66, pp. 2882–2896, Oct. 2023.
- [35] Y. Tong, J. Liu, H. Zhou, Z. Ju, and X. Zhang, "Adaptive tracking control of robotic manipulators with unknown kinematics and uncertain dynamics," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 4, pp. 5252–5269, Oct. 2024.
- [36] J. Fu, I. Burzo, E. Iovene, J. Zhao, G. Ferrigno, and E. De Momi, "Optimization-based variable impedance control of robotic manipulator for medical contact tasks," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–8, 2024.
- [37] X. Liu, Z. Liu, and P. Huang, "Stochastic optimal control for robot manipulation skill learning under time-varying uncertain environment," *IEEE Trans. Cybern.*, vol. 54, no. 4, pp. 2015–2025, Apr. 2024.
- [38] H. Chen and Y. Liu, "Robotic assembly automation using robust compliant control," *Robot. Comput. Integr. Manuf.*, vol. 29, no. 2, pp. 293–300, 2013.
- [39] T. Takagi and M. Sugeno, "Fuzzy identification of systems and its applications to modeling and control," *IEEE Trans. Syst., Man, Cybern.*, vol. 15, no. 1, pp. 116–132, Jan./Feb. 1985.
- [40] N. Wang, C. Chen, and A. D. Nuovo, "A framework of hybrid force/motion skills learning for robots," *IEEE Trans. Cogn. Develop. Syst.*, vol. 13, no. 1, pp. 162–170, Mar. 2021.



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